

Curriculum Vitae

Abinitha Gourabathina
PhD Student @ MIT EECS | MIT LIDS

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Research Interests

My research focuses on improving model reliability by leveraging information encoded within trained models. I study how internal representations can reveal model failures and guide methods for more robust, calibrated, and controllable behavior under diverse conditions.

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA 2024–2028 (Expected)
PhD, Electrical Engineering and Computer Science 2028
SM, Electrical Engineering and Computer Science 2026
GPA: 5.0/5.0

Princeton University, Princeton, NJ 2019–2023
BSE, Operations Research & Financial Engineering
Thesis: *What Seems to be the Problem? Stigmatizing Language in Patient Medical Notes*
Magna Cum Laude
Minors in Computer Science; Cognitive Science; Linguistics; Statistics & Machine Learning
GPA: 3.9/4.0

Professional Positions

MIT LIDS, Research Assistant 09/2024–Present
— Advisors: Prof. Collin Stultz & Prof. Marzyeh Ghassemi
— Computational Cardiovascular Research Group ([CCRG](#)) and [Healthy ML](#) research group
— Developing methods that use information encoded within trained models to improve reliability, with work on robustness under distribution shift, LLM reasoning and abstention, and model evaluation

IBM Research, AI Research Intern 06/2025–08/2025
— Mentor: Prasanna Sattigeri
— Developed the Query Misalignment framework for LLM abstention, reframing failures as models answering a different question from the one asked
— Proposed Trace Inversion, which reconstructs a query from the model's reasoning trace and uses query-reconstruction similarity to detect likely errors
— Work accepted to [ACL 2026 Main Conference \(Oral\)](#)

Bridgewater Associates, Investment Associate 09/2023–07/2024
— Studied macroeconomic investment strategies, focusing on systematizing quantitative strategies for sustainable equities portfolios

Bridgewater Associates, Investment Associate Intern 06/2022–08/2022

Carisk Partners, Data and Product Intern 03/2020–05/2022
— Built several machine-learning models to estimate recovery time for patients from lower income minority communities by utilizing clinical text data and adapting traditional NLTK approaches to medical jargon using ontology linking
— Conducted NLP-based data analysis of patient chatting platforms from the bottom-up, presenting findings to team of medical professionals on patient engagement

Vanderbilt University, REU Summer Research Intern 06/2021–08/2021
— Advisor: Bradley Malin, Mentor: Zhiyu Wan
— Developed game-theoretic framework to maximize the amount of COVID-19 data shared while maintaining HIPAA compliance and modeling the resulting privacy risk using deep learning models
— Conducted research project funded by the National Library of Medicine (NLM) and the National Science Foundation Research Experiences for Undergraduates (REU)

Publications

Peer-Reviewed Publications

- **Rank-Constrained Adaptation for Reliable Real-World Performance.**
Abinitha Gourabathina, Hyewon Jeong, Teya Bergamaschi, Marzyeh Ghassemi, Collin Stultz. **NeurIPS 2026 Main**. [paper] [code]
- **Poison-then-Hide: Finetuning-Activated Backdoor Attack on Pretrained Vision Encoders.**
Qixuan Jin, **Abinitha Gourabathina**, Vinith Menon Suriyakumar, Walter Gerych, Marzyeh Ghassemi. **NeurIPS 2026 Main**.
- **Sense and Sensitivity: Benchmarking LLM Clinical Triage Recommendations with Physician Experts.**
Abinitha Gourabathina, Haoran Zhang, Yueying Hao, Walter Gerych, Marzyeh Ghassemi. **Findings of EMNLP 2026**. [paper] [code]
- **Answering the Wrong Question: Reasoning Trace Inversion for Abstention in LLMs.**
Abinitha Gourabathina, Inkit Padhi, Manish Nagireddy, Subhajit Chaudhury, Prasanna Sattigeri. **ACL 2026 Main Conference (Oral)**. [paper] [code]
- **The Medium is the Message: How Non-Clinical Information Shapes Clinical Decisions in LLMs.**
Abinitha Gourabathina, Walter Gerych, Eileen Pan, Marzyeh Ghassemi. **FAccT 2025 (Oral)**. Association for Computing Machinery, 1805–1828. [paper] [code]
- **PanDa Game: Optimized Privacy-Preserving Publishing of Individual-Level Pandemic Data Based on a Game Theoretic Model.**
Abinitha Gourabathina, Zhiyu Wan, J. Thomas Brown, Chao Yan, Bradley A. Malin. **IEEE Transactions on NanoBioscience**, 22(4):808–817, 2023. [paper]
- **PanDa Game: Optimized Privacy-Preserving Publishing of Individual-Level Pandemic Data Based on a Game Theoretic Model.**
Abinitha Gourabathina, Zhiyu Wan, J. Thomas Brown, Chao Yan, Bradley A. Malin. **IEEE BIBM**, 2022. [paper]

Preprints

- **The MedPerturb Dataset: What Non-Content Perturbations Reveal About Human and Clinical LLM Decision Making.**
Abinitha Gourabathina, Yueying Hao, Walter Gerych, Marzyeh Ghassemi. 2025. arXiv:2506.17163. [paper] [website] [code]

Manuscripts Under Review

- **Subtyping with ConCEPT: Contrastive Clinical Embeddings from Patient Trajectories.**
Teya Bergamaschi, **Abinitha Gourabathina**, Collin Stultz. 2026.

Other Works

- **What Seems to be the Problem? Stigmatizing Language in Patient Medical Notes.**
Abinitha Gourabathina. Princeton University Senior Thesis, 2023. [paper]
- **Corpus Analysis of English Modifying Adverbs.**
Abinitha Gourabathina, Christiane Fellbaum. In T. Khan, M. Bapuji, & A. Satapathy (Eds.), *Alternative Horizons in Linguistics: A Festschrift in Honour of Prof. Panchanan Mohanty*. LINCOM GmbH, 2021.

Selected Academic Awards, Teaching, and Service

Best Independent Work Prize from Princeton University Center of Statistics and Machine Learning	2023
Sigma Xi Research Award	2023
Recipient of Eric F. S. Pai '83 Summer Research Grant	2022

Teaching

COS401: Machine Translation, Head Teaching Assistant	Spring 2023
— Taught lectures, organized practicum sections for bridging coding experience, and helped create problem sets	
— Course taught by Lecturer Srinivas Bangalore	
ORF245: Fundamentals of Statistics, Teaching Assistant	Spring 2021
— Hosted weekly office hours and graded problem sets	
— Course taught by Professor Matias Cattaneo	

Selected Talks & Presentations

Invited Talk, RBC Borealis	2026
ML Tea Talk, MIT	2025
FAccT '25 Main Conference Presentation, ACM	2025
Social and Ethical Responsibilities of Computing (SERC) Research Symposium, MIT	2025
IMES Seminar Series, MIT	2025

Departmental Service

Graduate Women in Course 6 (EECS) Executive Board, MIT	2025–Present
Laboratory of Information & Decision Systems (LIDS) Student Committee, MIT	2025–Present

Professional Service

Outreach Subchair, Machine Learning for Health (ML4H)	2026
Reviewer, TMLR	2026
Program Chair for 2026 LIDS Student Conference, MIT	2025–2026
Organizer of GW6 Research Summit, MIT	2025–Present
Reviewer for TS4H Workshop, NeurIPS '25	2025
Program Committee for AIES Conference, AIES '25	2025